

$$N_{C_2} = \frac{N(N-1)}{2}$$

, $w = 1/N$

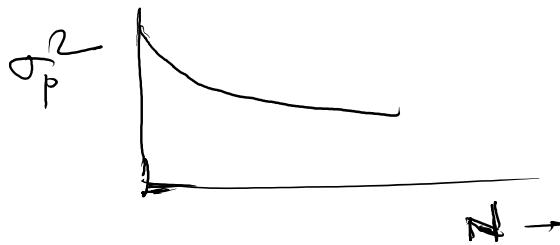
$$\sigma_p^2 = N w^2 \sigma^2 + 2 w^2 \rho \sigma^2 \times \frac{N(N-1)}{2}$$

$$= \frac{\sigma^2}{N} + \rho \sigma^2 \left(\frac{N-1}{N} \right)$$

$$= \frac{\sigma^2}{N} (1 + \rho(N-1)) = \sigma^2 \left(\rho + \frac{1-\rho}{N} \right)$$

$$N \rightarrow \infty, \quad \sigma_p^2 = \rho \sigma^2 \quad (\text{lower limit})$$

$$\rho = 0, \quad \sigma_p^2 = \sigma^2 / N$$



$$\text{Min } f(x)$$

s.t.

$$g_1(x) = c_1$$

$$g_2(x) = c_2$$



$$L(x, \lambda_1, \lambda_2) = \underline{f(x)} + \underset{\uparrow}{\lambda_1} (\underset{\uparrow}{g_1(x)} - \underline{c_1}) + \underset{\uparrow}{\lambda_2} (\underset{\uparrow}{g_2(x)} - \underline{c_2})$$

$$\frac{\partial L}{\partial x} = 0$$

$$\frac{\partial L}{\partial \lambda_1} = 0 \rightarrow g_1(x) - c_1 = 0$$

$$\frac{\partial L}{\partial \lambda_2} = 0 \rightarrow g_2(x) - c_2 = 0$$



↓
optimal
solution.

$$\frac{\partial}{\partial \underline{w}} (\underline{w}^T \underline{\mu}) = 0$$

$$y = \underline{w}^T \underline{\mu} = w_1 \mu_1 + w_2 \mu_2$$

$$\frac{\partial y}{\partial w_1} = \mu_1$$

$$\frac{\partial y}{\partial w_2} = \mu_2$$

$$\begin{pmatrix} \frac{\partial y}{\partial w_1} \\ \frac{\partial y}{\partial w_2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

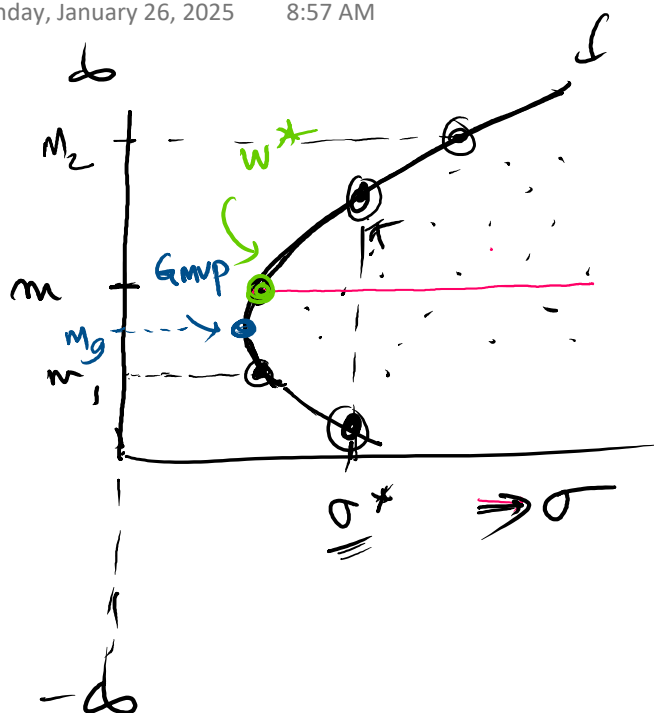
$\underline{\mu} = 0$

$$y = \frac{1}{2} \mathbf{w}^T \underline{\underline{\Omega}} \mathbf{w} = \frac{1}{2} \sigma_p^2 = \frac{1}{2} \left(w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2\rho w_1 w_2 \sigma_1 \sigma_2 \right)$$

$$\rightarrow \frac{\partial y}{\partial w_1} = \frac{1}{2} (2w_1 \sigma_1^2 + 2\rho w_2 \sigma_1 \sigma_2) = \underline{w_1 \sigma_1^2 + \rho w_2 \sigma_1 \sigma_2}$$

$$\rightarrow \frac{\partial y}{\partial w_2} = \underline{w_2 \sigma_2^2 + \rho w_1 \sigma_1 \sigma_2}$$

$$\nabla y = \frac{\partial y}{\partial \mathbf{w}} = \begin{pmatrix} \frac{\partial y}{\partial w_1} \\ \frac{\partial y}{\partial w_2} \end{pmatrix} = \begin{pmatrix} w_1 \sigma_1^2 + \rho w_2 \sigma_1 \sigma_2 \\ w_2 \sigma_2^2 + \rho w_1 \sigma_1 \sigma_2 \end{pmatrix} = \begin{pmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \underline{\underline{\Omega}} \mathbf{w}$$



Minimum
variance
frontier

μ ↑

$$\min \frac{1}{2} W^T \Sigma W$$

RFA (1)

w_0

Risky Asset (n)

$$W = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \neq$$

$$\sigma_p^2 = \underbrace{w_0^2 \sigma_0^2 + \left(w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + \dots + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2 + \dots \right)}_{(W^T \Omega W)}$$

$$\mu_p = w_0 r_f + w_1 \mu_1 + w_2 \mu_2 + \dots + w_n \mu_n$$

$$= (1 - w_1 - w_2 - \dots - w_n) r_f + w_1 \mu_1 + w_2 \mu_2 + \dots + w_n \mu_n$$

$$= r_f + w_1 (\mu_1 - r_f) + w_2 (\mu_2 - r_f) + \dots + w_n (\mu_n - r_f)$$

$$\mu_p = r_f + W^T (\mu - r_f \mathbf{1}) = m$$

$$(w_1 \ w_2 \ \dots \ w_n) \left[\begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix} - r_f \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \right]$$